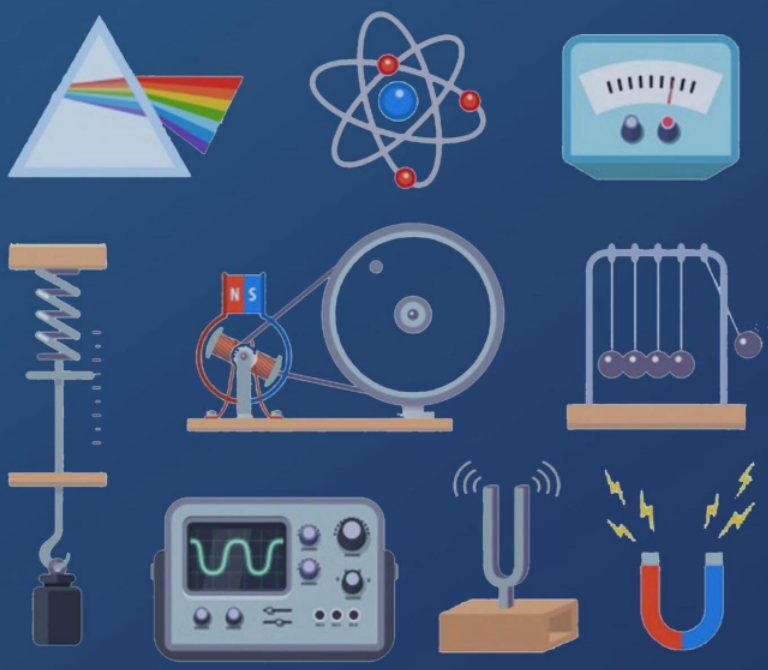
PHYSICS

Center of Mass



SHEET

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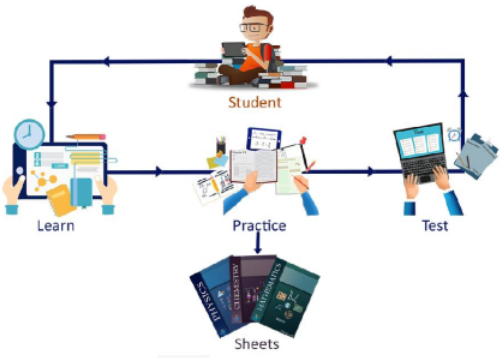
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Locating centre of mass -Discrete Bodies		19,20		
2	Locating centre of mass -Continuous Bodies	1-4	21	1-3	25-27
3	Motion of Centre of mass	5-11	7,18,22	4-9	1
4	Linear Momentum	12-16			
5	Linear momentum Conservation - Part 1	17-21	1	10	14-17
6	Linear momentum Conservation - Part 2	22,23	8	11-13	2,9,18-20, 24,28
7	Impulse	24-28		14,15	
8	Head on collision - Part 1	29,30	9,10,13, 14,23	16,17	3,10
9	Head on collision - Part 2	31,32	2,11,12	18-21	4-6,29
10	Oblique collision			22-27	7,30
11	Centre of mass Frame			28	11
12	Variable mass	33			21-23
13	COM - How things are derived	34,35			
14	After Full Chapter	36,37	3-6, 15-17,24		8,12,13, 31-34

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Locating centre of mass -Continuous Bodies

Q1. The centre of mass of a body .....

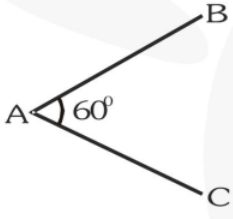
(A) Lies always outside the body.

(B) May lie within, outside or on the surface of the body.

(C) Lies always inside the body.

(D) Lies always on the surface of the body.

Q2. A uniform wire of length ℓ is bent into the shape of 'V' as shown. The distance of its centre of mass from the vertex A is :-



(A) $\ell/2$ (B) $\frac{\ell\sqrt{3}}{4}$

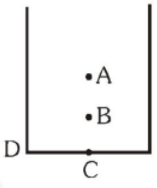
(C) $\frac{\ell\sqrt{3}}{8}$ (D) None of these

Q3. The centre of mass of a non uniform rod of length L whose mass per unit length varies as $\lambda = kx^2/L$ (where k is a constant and x is the distance measured from one end) is at the following distance from the same end :-

(A) 3L/4 (B) L/4

(C) 2L/3 (D) L/3

Q4. A thick uniform wire is bent into the shape of the letter "U" as shown. Which point indicates the location of the center of mass of this wire? A is the midpoint of the line joining mid points of two parallel sides of 'U' shaped wire.

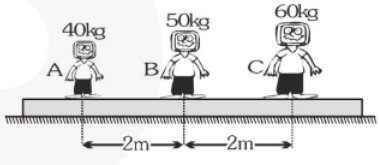


(A) D (B) A

(C) B (D) C

Motion of Centre of mass

Q5. Three man A, B & C of mass 40 kg, 50 kg & 60 kg are standing on a plank of mass 90 kg, which is kept on a smooth horizontal plane. If A & C exchange their positions then mass B will shift :-



(A) 1/3 m towards left

(B) 1/3m towards right

(C) will not move w.r.t. ground

(D) 5/3 m towards left

Q6. An isolated particle of mass m is moving in horizontal plane (x–y), along the x–axis, at $y = 0$. It suddenly explodes into two fragment of masses $\frac{m}{4}$ and $\frac{3m}{4}$. An instant later, the smaller fragment is at $y = +15$ cm. The larger fragment at this instant is at :-

(A) $y = -5$ cm (B) $y = +20$ cm

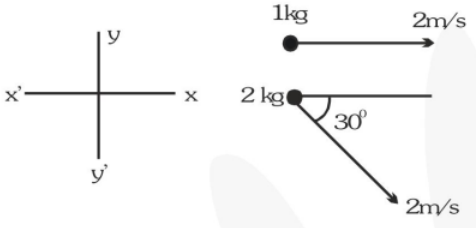
(C) $y = +5$ cm (D) $y = -20$ cm

Q7. Two particles A and B initially at rest, move towards each other under the mutual force of attraction. At the instant when the speed of A is v and the speed of B is 2v, the speed of the centre of mass of the system is :-

(A) 3v (B) v

(C) 1.5v (D) zero

Q8. The velocity of centre of mass of the system as shown in the figure :-



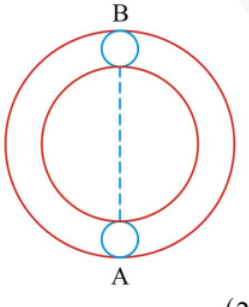
(A) $\left(\frac{2-2\sqrt{3}}{3}\right)\hat{i}-\frac{1}{3}\hat{j}$

(B) $\left(\frac{2+2\sqrt{3}}{3}\right)\hat{i}-\frac{2}{3}\hat{j}$

(C) $4\hat{i}$

(D) None of these

Q9. Two equal spheres A and B lie on a smooth horizontal circular groove at opposite ends of a diameter. At time t =0, A is projected along the groove and it first impinges on B at time t =T_1 and again at timet = T_2. If e is the coefficient of restitution the ratio T_2/T_1 is



(A) $\frac{2}{e}$ (B) $\frac{(2+e)}{2}$

(C) $\frac{2(e+1)}{e}$ (D) $\frac{(2+e)}{e}$

Q10. Sachin (55 kg) and Kapil (65 kg) are sitting at the two ends of a boat at rest in still water. The boat weighs 100 kg and is 3.0 m along. Sachin walks down to Kapil and shakes hand. The boat gets displaced by :-

(A) zero m (B) 0.75 m

(C) 3.0 m (D) 2.3 m

Q11. The velocity of centre of mass in absence of external force is :-

(A) constant (B) zero

(C) increases (D) decreases

Linear Momentum

Q12. Two bodies with masses M_1 and M_2 have equal kinetic energies. If P_1 and P_2 are their respective momenta, then P_1/P_2 is equal to ?

(A) $M_1 : M_2$ (B) $M_1^2 : M_2^2$

(C) $M_2 : M_1$ (D) $\sqrt{M_1} : \sqrt{M_2}$

Q13. A cricket ball of mass 150 gm has an initial velocity $\vec{u} = (3\hat{i}+4\hat{j})$ m / s and a final velocity $\vec{v} = -(3\hat{i}+4\hat{j})$ m / s after being hit. Find the change in momentum (in kg m/s) :-

(A) Zero

(B) $-(0.45\hat{i}+0.6\hat{j})$

(C) $-(0.9\hat{i}+1.2\hat{j})$

(D) $-5(\hat{i}+\hat{j})$

Q14. The velocity of a body of mass 2 kg as a function of t is given by $v(t) = 2t\,\hat{i}+t^2\hat{j}$. Find the momentum at time t = 2 sec.

(A) $(8\,\hat{i}+8\hat{j})$ kg m / s

(B) $2(4\,\hat{i}+4\hat{j})$ kg m / s

(C) $2[2(2)\,\hat{i}+(2)^2\hat{j}]$ kg m / s

(D) All of the above

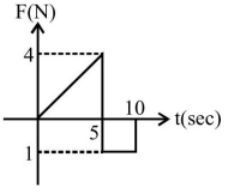
Q15. A force time graph for the motion of a body is shown. The change in momentum of the body between zero and 10 sec is :-

(A) 15 kgm/sec

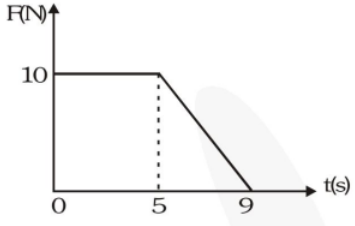
(B) 4 kgm/sec

(C) 6 kgm/sec

(D) 5 kgm/sec



Q16. A body of mass 4 kg is acted on by a force which varies as shown in the graph below. The momentum acquired is -



(A) 280 N–s (B) 140 N–s

(C) 70 N–s (D) 210 N–s

Linear momentum Conservation - Part 1

Q17. A bomb of 12 kg explodes into two pieces of masses 4 kg and 8 kg. The velocity of 8 kg mass is 6 m/sec. The kinetic energy of the other mass is :-

(A) 48 J (B) 32 J

(C) 24 J (D) 288 J

Q18. Two particles A and B start moving due to their mutual interaction only. If at any time 't', $\vec{a}_A$ and $\vec{a}_B$ are their respective accelerations, $\vec{v}_A$ and $\vec{v}_B$ are their respective velocities, and upto that time W_A and W_B are the work done on A and B respectively by the mutual force, m_A and m_B are their masses respectively, then which of the following is always correct.

(A) $\vec{v}_A + \vec{v}_B = 0$

(B) $m_A\vec{v}_A + m_B\vec{v}_B = 0$

(C) $W_A + W_B = 0$

(D) $\vec{a}_A + \vec{a}_B = 0$

Q19. Two particles A and B initially at rest move towards each other under a mutual force of attraction. At the instant when the speed of A is v and the speed of B is 2v, the speed of centre of mass of the system is :-

(A) Zero (B) v

(C) 1.5v (D) 3v

Q20. A rifle of mass M is initially at rest but free to recoil. It fires a bullet of mass m and muzzle velocity v. After firing, the velocity of the rifle (relative to the ground) is:-

(A) –mv (B) –Mv/m

(C) –mv/M (D) –v

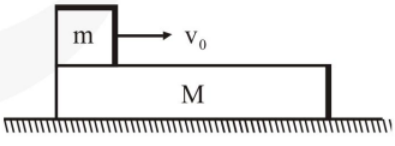
Q21. A gun fires a bullet of mass 50 g with a velocity of 30 $msec^{-1}$. Because of this the gun is pushed back with a velocity of 1 m/sec^{-1}. The mass of the gun is:-

(A) 15 kg (B) 30 kg

(C) 1.5 kg (D) 20 kg

Linear momentum Conservation - Part 2

Q22. The coefficient of friction between the block and plank is μ and ground is smooth. The value of μ is such that block becomes stationary with respect to plank before it reaches the other end. Then which of the following statement is **incorrect.**



(A) The work done by friction on the block is negative.

(B) The work done by friction on the plank is positive .

(C) The net work done by friction is negative.

(D) Net work done by the friction is zero.

Q23. A bomb of mass 3m kg explodes into two pieces of mass m kg and 2m kg. If the velocity of m kg mass is 16 m/s, the total kinetic energy released in the explosion is:-

(A) 192m J (B) 96m J

(C) 384m J (D) 768m J

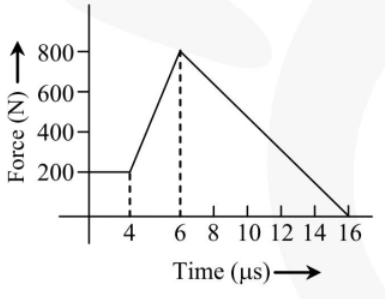
Impulse

Q24. An impulse $\vec{I}$ changes the velocity of a particle from $\vec{v}_1$ to $\vec{v}_2$. Kinetic energy gained by the particle is :-

(A) $\frac{1}{2}\vec{I}\cdot(\vec{v}_1+\vec{v}_2)$ (B) $\frac{1}{2}\vec{I}\cdot(\vec{v}_1-\vec{v}_2)$

(C) $\vec{I}\cdot(\vec{v}_1-\vec{v}_2)$ (D) $\vec{I}\cdot(\vec{v}_1+\vec{v}_2)$

Q25. The magnitude of force (in Newtons) acting on a body varies with time (in micro second) as shown in the figure. The magnitude of total impulse of the force on the body from t = 4 μs to t = 16 μs is -



(A) 5×10^{-2} Ns (B) 5×10^{-3} Ns

(C) 5×10^{-4} Ns (D) 5×10^{-6} Ns

Q26. A ball of mass 1 kg drops vertically on to the floor with a speed of 25 m/s. It rebounds with an initial velocity of 10 m/s. What impulse acts on the ball during contact?

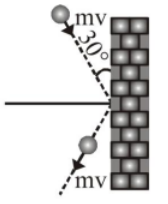
(A) 35kg m/s downwards

(B) 35 kg m/s upwards

(C) 30 kg m/s downwards

(D) 30kg m/s upwards

Q27. An iron ball of mass 100 g having velocity 10 m/s collides with a wall at an angle 30° and rebounds with the same angle. If the period of contact between the ball and wall is 0.1s, then the force experienced by the ball is:-



(A) 100 N (B) 10 N

(C) 0.1 N (D) 1.0 N

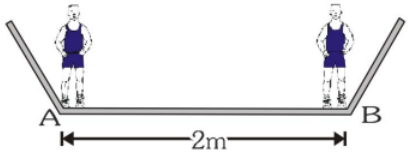
Q28. Two block of masses 10 kg and 4kg are connected by a spring of negligible mass and placed on a frictionless horizontal surface. An impulse gives a velocity of 14m/s to the heavier block in the direction of the lighter block. The velocity of the centre of mass is:-

(A) 30 m/s (B) 20 m/s

(C) 10 m/s (D) 5 m/s

Head on collision - Part 1

Q29. Two persons A and B of weight 80 kg and 50 kg respectively are standing at opposite ends of a boat of mass 70 kg and length 2m at rest. When they interchange their positions then displacement of the centre of mass of the boat will be :-



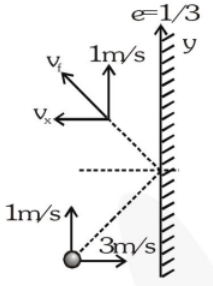
(A) 60 cm towards left

(B) 30 cm towards right

(C) 30 cm towards left

(D) stationary

Q30. A smooth sphere is moving on a horizontal surface with velocity vector $3\hat{i}+\hat{j}$ imme diately before it hits a vertical wall. The wall is parallel to the vector $\hat{j}$ and the coefficient of restitution between the wall and sphere is $\frac{1}{3}$. The velocity vector of the sphere after it hits the wall is :-



(A) $\hat{i}+\hat{j}$ (B) $3\hat{i}-\frac{1}{3}\hat{j}$

(C) $-\hat{i}+\hat{j}$ (D) $\hat{i}-\hat{j}$

Head on collision - Part 2

Q31. A ball hits the floor and rebounds after an inelastic collision. In this case :-

(A) the momentum of the ball just after the collision is the same as that just before the collision

(B) the mechanical energy of the ball remains the same in the collision

(C) the total momentum of the ball and the earth is conserved

(D) the total energy of the ball and the earth is conserved

Q32. A ball of mass 2m impinges directly on a ball of mass m, which is at rest. If the velocity with which the larger ball impinges be equal to the velocity of the smaller mass after impact then the coefficient of restitution :-

(A) $\frac{1}{3}$ (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) $\frac{2}{5}$

Variable mass

Q33. If in a rocket relative velocity v of exhaust gas is constant and mass flow rate is $\frac{dm}{dt}$ then net force on the rocket is :-

(A) $v\frac{dm}{dt}$ (B) $m\frac{dv}{dt}$ (C) 0 (D) ∞

COM - How things are derived

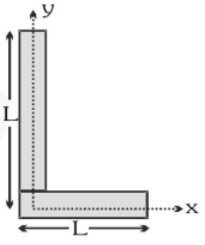
Q34. Centre of mass of two uniform rods of same length but made up of different materials & kept as shown, if the meeting point is the origin of co–ordinates :-

(A) (L/2,L/2)

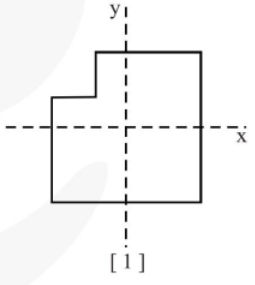
(B) (2L/3,L/2)

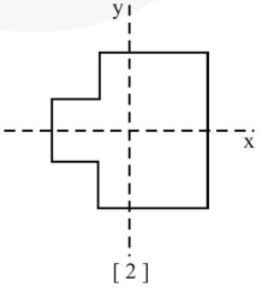
(C) (L/3,L/3)

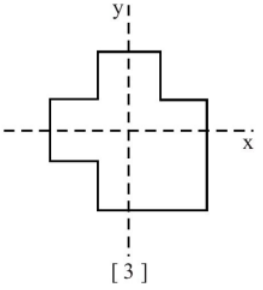
(D) (L/3,L/6)



Q35. A machinist starts with three identical square plates but cuts one corner from one of them, two corners from the second, and three corners from the third. Rank the three plates according to the x-coordinate of their centers of mass, from smallest to largest.







(A) 3, 1, 2 (B) 1, 3, 2

(C) 3, 2, 1 (D) 1 and 3 tie, then 2

After Full Chapter

Q36. As shown, a big box of mass M is resting on a horizontal smooth floor. On the bottom of the box there is a small block of mass m. The block is given an initial speed v_0 relative to the floor, and starts to bounce back and forth between the two walls of the box. Find the final speed of the box when the block has finally come to rest in the box :-



(A) 0 (B) v_0

(C) $\frac{m}{M}v_0$ (D) $\frac{m}{m+M}v_0$

Q37. A body of mass M moves in outer space with velocity v. It is desired to break the body into two parts so that the mass of one part is one–tenth of the total mass. After the explosion, the heavier part comes to rest while the lighter part continues to move in the original direction of motion. The velocity of the small part will be :

(A) v (B) $\left(\frac{v}{2}\right)$

(C) $\left(\frac{v}{10}\right)$ (D) 10 v

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A bag of mass M hangs by a long thread and a bullet (mass m) comes horizontally with velocity v and gets caught in the bag. Then for the combined system (bag + bullet) :-

(A) Momentum is $mMv/(M + m)$

(B) kinetic energy is $(1/2)\,Mv^2$

(C) Momentum is mv

(D) kinetic energy is $m^2v^2/2(M + m)$

Q2. The Fig. shows a string of equally placed beads of mass m, separated by distance d. The beads are free to slide without friction on a thin wire. A constant force F acts on the first bead initially at rest till it makes collision with the second bead. The second bead then collides with the third and so on. Suppose that all collisions are elastic :



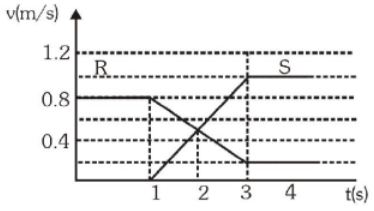
(A) Speed of the first bead immediately before and immediately after its collision with the second bead is $\sqrt{\frac{2Fd}{m}}$ and zero respectively.

(B) Speed of the first bead immediately before and immediately after its collision with the second bead is $\sqrt{\frac{2Fd}{m}}$ and $\frac{1}{2}\sqrt{\frac{2Fd}{m}}$ respectively.

(C) Speed of the second bead immediately after its collision with third bead is zero.

(D) The average speed of the first bead is $\frac{1}{2}\sqrt{\frac{2Fd}{m}}$.

Q3. The diagram shows the velocity–time graph for two masses R and S that collided elastically. Which of the following statements is true ?



I. R and S moved in the same direction after the collision

II. The velocities of R and S were equal at the mid time of the collision.

III. The mass of R was greater than mass of S.

(A) I only (B) II only

(C) I and II only (D) I, II and III

Q4. Two charges moving under their only own mutual attraction separated by large distance initially. Then choose the correct statement (s) :

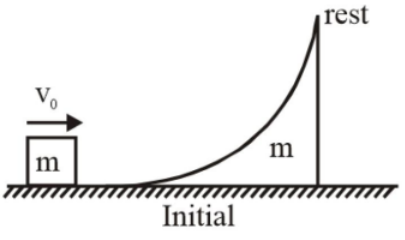
(A) If both are free, mechanical energy is conserved.

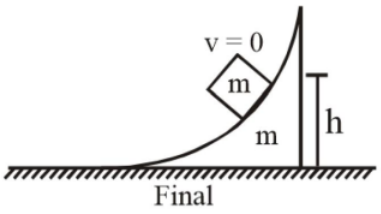
(B) If one is fixed and other is free, mechanical energy is conserved.

(C) If one is fixed and other is free, momentum is conserved.

(D) If both are free momentum is conserved.

Q5. In the arrangement shown, horizontal surface is smooth, but friction is present between the block and the surface of the wedge. Block is given velocity v_0 at $t = 0$. After achieving height 'h' on the wedge, block comes to rest with respect to wedge at $t = t_0$. Then from $t = 0$ to $t = t_0$:-





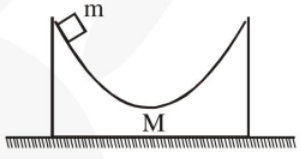
(A) Work done by friction on the block is negative

(B) Work done by friction on the wedge is negative

(C) Work done by block on the wedge is positive

(D) Work done by wedge on the block is positive

Q6. Figure shows a wedge on which a small block is released from rest. All the surfaces are smooth system comprises of wedge and blocks. Mark the correct statement(s) regarding motion of block on wedge till block attains maximum height on wedge.



(A) Acceleration of centre of mass of system is initially vertically down then vertically up.

(B) Initially centre of mass moves down and then up.

(C) At the maximum height block and wedge move with common velocity.

(D) Centre of mass of wedge moves towards left then right

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q7. A particle of mass 1kg has velocity $\vec{v}_1 = (2t)\hat{i}$ and another particle of mass 2 kg has velocity $\vec{v}_2 = (t^2)\hat{j}$.

Column I	Column II
(A) Net force on centre of mass at 2 s	(P) $\frac{20}{9}$ unit
(B) Velocity of centre of mass at 2s	(Q) $\sqrt{68}$ unit
(C) Displacement of centre of mass in 2s	(R) $\frac{\sqrt{80}}{3}$ unit
	(S) None

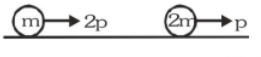
Q8. If net force on a system of particles is zero, then

Column I	Column II
(A) Acceleration of centre of mass	(P) Constant
(B) Velocity of centre of mass	(Q) Zero
(C) Momentum of centre of mass	(R) May be zero
(D) Velocity of an individual particle of the system	(S) May be constant

Q9. A particle of mass m, kinetic energy K and momentum p collides head on elastically with another particle of mass 2 m at rest. After collision :

Column I	Column II
(A) Momentum of first particle	(P) 3/4 p
(B) Momentum of second particle	(Q) – K/9
(C) Kinetic energy of first particle	(R) – p/3
(D) Kinetic energy of second particle	(S) $\frac{8K}{9}$
	(T) None

Q10. Two balls of mass m and 2m each have momentum 2p and p in the direction shown in figure. During collision they exert an impulse of magnitude p on each other.

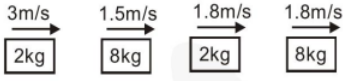
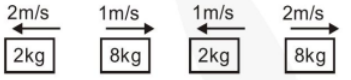
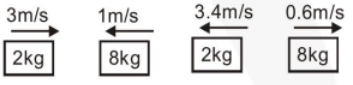


Column I	Column II
(A) After collision momentum of m	(P) 2p
(B) After collision momentum of 2m	(Q) p
(C) Coefficient of restitution between them	(R) 1
	(S) None

Q11.

Column I	Column II
(A) Elastic collision	(P) KE is conserved
(B) Inelastic collision	(Q) KE after collision = KE before collision
(C) Perfectly inelastic collision	(R) KE after collision ≠ KE before collision
	(S) Particles stick after collision
	(T) Linear momentum is conserved
	(U) Relative velocity of separation after collision is zero

Q12. In Column-I, 4 situations are depicted and in column-II, 4 possible kinds of collision are listed. Match the situation with type of collision.

Column-I (Before)	Column-II (After)
(A) 	(P) Elastic
(B) 	(Q) Perfectly Inelastic
(C) 	(R) Partially elastic
	(S) Collision is not possible

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

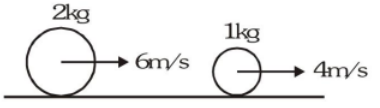
Paragraph # 1 (Ques. 13 to 14)

When two bodies collide normally they exert equal and opposite impulses on each other.

Impulse = change in linear momentum. Coefficient of restitution between two bodies is given by :-

$$e = \frac{|\text{Relative velocity of separation}|}{|\text{Relative velocity of approach}|} = 1,$$

for elastic collision



Two bodies collide as shown in figure. During collision they exert impulse of magnitude J on each other.

Q13. If the collision is elastic, the value of J is ..... N–s :-

(A) 10/3 (B) 5/4

(C) 8/3 (D) 3/2

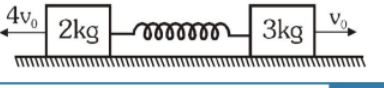
Q14. For what values of J (in N–s) the 2 kg block will change its direction of velocity :-

(A) $J < 12$ (B) $J > 12$

(C) $J < 10$ (D) $J > 10$

Paragraph # 2 (Ques. 15 to 17)

2 kg and 3 kg blocks are placed on a smooth horizontal surface and connected by spring which is unstretched initially. The blocks are imparted velocities as shown in the figure.



Q15. The maximum energy stored in the spring in the subsequent motion will be

(A) $5v_0^2$ (B) $15v_0^2$

(C) zero (D) $10v_0^2$

Q16. Maximum speed of 3 kg block in the subsequent motion will be

(A) v_0 (B) $2v_0$

(C) $3v_0$ (D) $4v_0$

Q17. Maximum speed of 2 kg block in the subsequent motion will be

(A) v_0 (B) $2v_0$

(C) $3v_0$ (D) $4v_0$

NUMERICAL TYPE

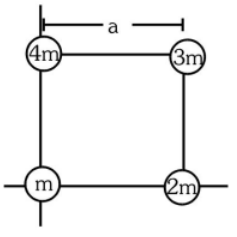
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q18. The figure shows the positions and velocities of two particles. If the particles move under the mutual attraction of each other, then find the position of centre of mass at t = 1 s.



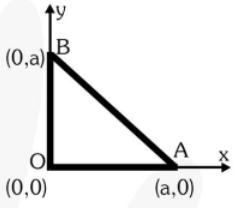
SUBJECTIVE TYPE

Q19. Four particles of masses m, 2m, 3m, 4m are placed at corners of a square of side 'a' as shown in fig. Find out coordinates of centre of mass.

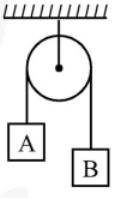


Q20. A rigid body consists of a 3 kg mass connected to a 2 kg mass by a massless rod. The 3kg mass is located at $\vec{r}_1 = (2\hat{i} + 5\hat{j})$ m and the 2 kg mass at $\vec{r}_2 = (4\hat{i} + 2\hat{j})$ m. Find the length of rod and the coordinates of the centre of mass.

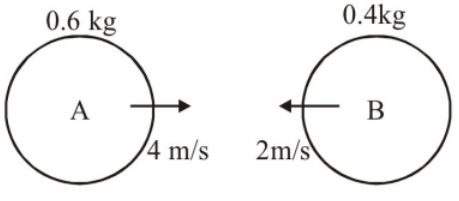
Q21. Three rods of the same mass are placed as shown in the figure. Calculate the coordinates of the centre of mass of the system.



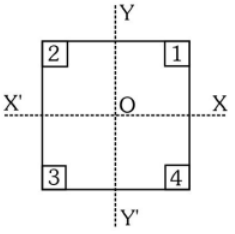
Q22. In the arrangement shown in the figure, $m_A = 2$ kg and $m_B = 1$ kg. String is light and inextensible. Find the acceleration of centre of mass of both the blocks. Neglect friction everywhere.



Q23. The velocities of two steel balls before impact are shown. If after head on impact the velocity of ball B is observed to be 3 m/s to the right, the coefficient of restitution is-



Q24. Figure shows a uniform square plate from which four identical squares at the corners will be removed.



(i) Where is the centre of mass of the plate originally.

(ii) Where is C.M. after square 1 is removed.

(iii) Where is C.M. after squares 1 and 2 removed.

(iv) Where is C.M. after squares 1 and 3 are removed.

(v) Where is C.M. after squares 1, 2 and 3 are removed.

(vi) Where is C.M. after all the four squares are removed.

Give Answers in terms of quadrants and axes.

Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. A piece of paper (shown in figure1) is in form of a square. Two corners of this square are folded to make it appear like figure.2 . Both corners are put together at centre of square 'O'. If O is taken to be (0,0), the centre of mass of new system will be at :

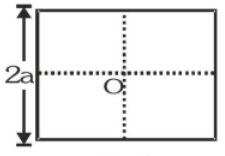


Fig.1

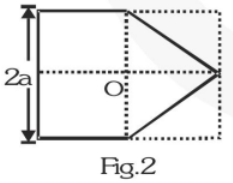
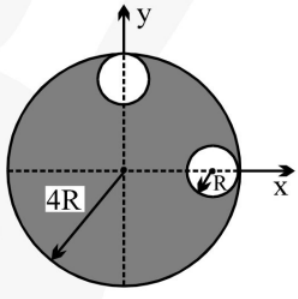


Fig.2

(A) $\left(-\frac{a}{8}, 0\right)$ (B) $\left(-\frac{a}{6}, 0\right)$

(C) $\left(\frac{a}{12}, 0\right)$ (D) $\left(-\frac{a}{12}, 0\right)$

Q2. From the circular disc of radius 4R two small disc of radius R are cut off. The centre of mass of the new structure will be at :



(A) $\frac{R}{5}\hat{i}+\frac{R}{5}\hat{j}$ (B) $-\frac{R}{5}\hat{i}+\frac{R}{5}\hat{j}$

(C) $\frac{-3R}{14}(\hat{i}+\hat{j})$ (D) None of these

Q3. If linear density of a rod of length 3 m varies as $\lambda = 2 + x$, then the position of the centre of gravity of the rod is:-

(A) $\frac{7}{2}$m (B) $\frac{12}{7}$m

(C) $\frac{10}{7}$m (D) $\frac{9}{7}$m

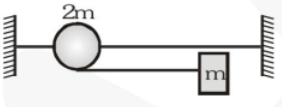
Q4. Considering a system having two masses m_1 and m_2 in which first mass is pushed towards centre of mass by a distance a, the distance required to be moved for second mass to keep centre of mass at same position is :-



(A) $\frac{m_1}{m_2}a$ (B) $\frac{m_1m_2}{a}$

(C) $\frac{m_2}{m_1}a$ (D) $\left(\frac{m_2m_1}{m_1+m_2}\right)a$

Q5. A bead can slide on a smooth straight wire and a particle of mass m attached to the bead by a light string of length L. The particle is held in contact with the wire and with the string taut and is then let fall. If the bead has mass 2m then when the string makes an angle θ with the wire, the bead will have slipped a distance.



(A) $L(1-\cos\theta)$ (B) $\left(\frac{L}{2}\right)(1-\cos\theta)$

(C) $\left(\frac{L}{3}\right)(1-\cos\theta)$ (D) $\left(\frac{L}{6}\right)(1-\cos\theta)$

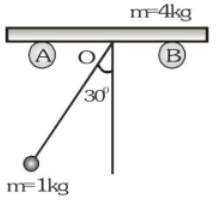
Q6. A ball of mass 1 kg is suspended by an inextensible string 1 m long attached to a point O of a smooth horizontal bar resting on a fixed smooth supports A and B. The ball is released from rest from the position when the string makes an angle of 30° with the vertical. The mass of the bar is 4 kg. The displacement in meters of the bar when the string makes the maximum angle on the other side of the vertical is :-

(A) 0

(B) 0.2

(C) 0.25

(D) 0.5



Q7. An open water tight railway wagon of mass 5 x 10^3 kg coasts at an initial velocity 1.2 m/s without friction on a railway track. Rain drops fall vertically downwards into the wagon. The velocity of the wagon after it has collected 10^3 kg of water will be :-

(A) 0.5 m/s (B) 2m/s

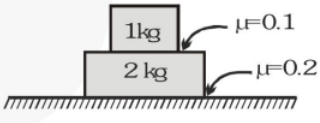
(C) 1 m/s (D) 1.5 m/s

Q8. Three interacting particles of masses 100 g, 200 g and 400 g each have a velocity of 20 m/s magnitude along the positive direction of x-axis, y-axis and z-axis. Due to force of interaction the third particle stops moving. The velocity of the second particle is $\left(10\hat{j}+5\hat{k}\right)$. What is the velocity of the first particle?

(A) $20\hat{i}+20\hat{j}+70\hat{k}$ (B) $10\hat{i}+20\hat{j}+8\hat{k}$

(C) $30\hat{i}+10\hat{j}+7\hat{k}$ (D) $15\hat{i}+5\hat{j}+60\hat{k}$

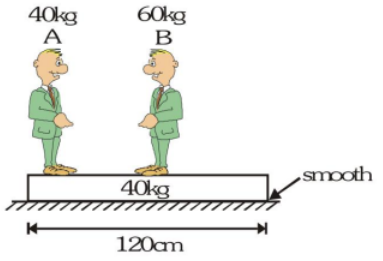
Q9. If both the blocks as shown in the given arrangement are given together a horizontal velocity towards right. If a_{cm} be the subsequent acceleration of the centre of mass of the system of blocks then a_{cm} equals :-



(A) 0 m/s² (B) $\frac{5}{3}$ m/s²

(C) $\frac{7}{3}$ m/s² (D) 2 m/s²

Q10. Two men 'A' and 'B' are standing on a plank. 'B' is at the middle of the plank and 'A' is at the left end of the plank. Lower surface of the plank is smooth. System is initially at rest and masses are as shown in figure. A and B starts moving such that the position of 'B' remains fixed with respect to ground then 'A' meets 'B'. Then the point where A meets B is located at :-



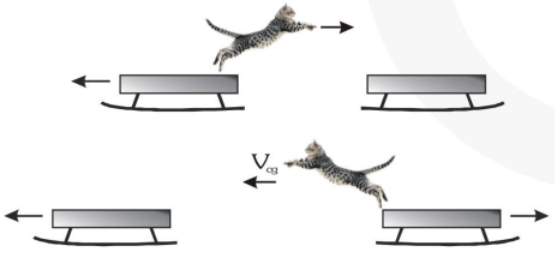
(A) the middle of the plank

(B) 30 cm from the left end of the plank

(C) the right end of the plank

(D) None of these

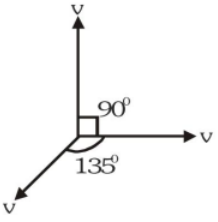
Q11. A strange cat with a mass m_c is sitting at rest on the left sled of a pair of identical sleds. The sleds have mass m_s and sit on frictionless ice. Suddenly, the cat leaps to the right sled, traveling with a horizontal speed v_{cg} measured with respect to the ground. The instant the cat reaches the right sled, it turns around and leaps back to the left sled. The horizontal component of the cat's speed is again v_{cg} measured with respect to the ground. The final speed of right sled in terms of the masses of the cat and sleds and the cat's leaping speed is (The cat remains on the left sled after its return) :-



(A) $\frac{2m_c v_{cg}}{m_s + m_c}$ (B) $\frac{2m_c v_{cg}}{m_s}$

(C) $\frac{m_c v_{cg}}{m_s + m_c}$ (D) $\frac{m_c v_{cg}}{m_s}$

Q12. A particle of mass 4m which is at rest explodes into four equal fragments. All 4 fragments scattered in the same horizontal plane. Three fragments are found to move with velocity v each as shown in the fig. The total energy released in the process of explosion is :-



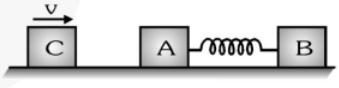
(A) $mv^2(3-\sqrt{2})$

(B) $mv^2(3-\sqrt{2})/2$

(C) $2mv^2$

(D $mv^2(1+\sqrt{2})/2$

Q13. Two blocks A and B each of mass m, are connected by a massless spring of natural length L and spring constant k. The blocks are initially resting on a smooth horizontal floor with the spring at its natural length, as shown in fig. A third identical block C, also of mass m, moves on the floor with a speed v along the line joining A and B, and collides elastically with A. Then :-



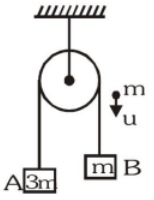
(A) the kinetic energy of the A–B system, at maximum compression of the spring, is zero

(B) the kinetic energy of the A–B system, at maximum compression of the spring, is $\frac{mv^2}{4}$

(C) the maximum compression of the spring is $v\sqrt{\left(\frac{m}{k}\right)}$

(D) the maximum compression of the spring is $2v\sqrt{\frac{m}{2k}}$

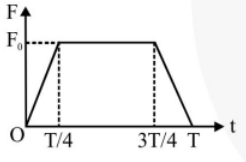
Q14. A system of two blocks A and B are connected by an inextensible massless strings as shown. The pulley is massless and frictionless. Initially the system is at rest when, a bullet of mass 'm' moving with a velocity 'u' as shown hits the block 'B' and gets embedded into it. The impulse imparted by tension force to the block of mass 3m is-



(A) $\frac{5mu}{4}$ (B) $\frac{4mu}{5}$

(C) $\frac{2mu}{5}$ (D) $\frac{3mu}{5}$

Q15. A particle of mass m moving with a velocity u makes an elastic one dimensional collision with a stationary particle of mass m establishing a contact with it for extremely small time T. Their force of contact increases from zero to F_0 linearly in time T/4, remains constant for a further time T/2 and decreases linearly from F_0 to zero in further time T/4 as shown. The magnitude possessed by F_0 is :-



(A) $\frac{mu}{T}$ (B) $\frac{2mu}{T}$

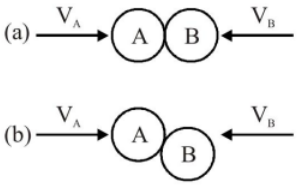
(C) $\frac{4mu}{3T}$ (D) $\frac{3mu}{4T}$

Q16. Two balls of same mass are dropped from the same height h, on to the floor. The first ball bounces to a height h/4 ,after the collision & the second ball to a height h/16. The impulse applied by the first & second ball on the floor are I_1 and I_2 respectively. Then :-

(A) $5I_1 = 6I_2$ (B) $6I_1 = 5I_2$

(C) $I_1 = 2I_2$ (D) $2I_1 = I_2$

Q17. Two bodies, A and B, collide as shown in figures a and b below :



Circle the true statement :

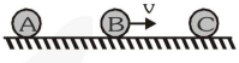
(A) They exert equal and opposite forces on each other in (a) but not in (b)

(B) They exert equal and opposite force on each other in (b) but not in (a)

(C) They exert equal and opposite force on each other in both (a) and (b)

(D) The forces are equal and opposite to each other in (a), but only the components of the forces parallel to the velocities are equal in (b).

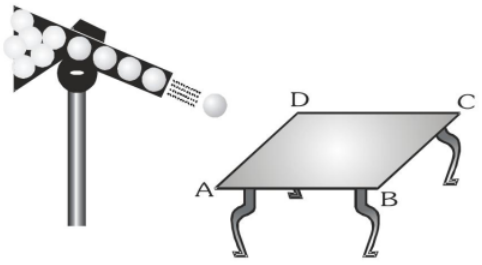
Q18. Three balls A, B and C ($m_A = m_C = 4m_B$) are placed on a smooth horizontal surface. Ball B collides with ball C with an initial velocity v as shown in the figure. Total number of collisions between the balls will be (All collisions are elastic) :-



(A) One (B) Two

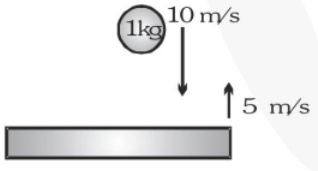
(C) Three (D) Four

Q19. A gun which fires small balls of mass 20 gm is firing 20 balls per second on the smooth horizontal table surface ABCD. If the collision is perfectly elastic and balls are striking at the centre of table with a speed 5 m/s at an angle of 60° with the vertical just before collision, then force exerted by one of the leg on ground is (assume total weight of the table is 0.2 kg and g = 10 m/s²) :-



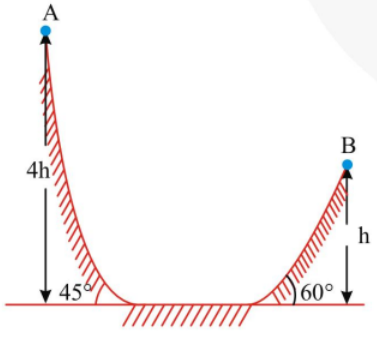
(A) 0.5 N (B) 1 N
(C) 0.25 N (D) 0.75 N

Q20. A ball of mass 1 kg strikes a heavy platform elastically, moving upwards with a velocity of 5m/s. The speed of the ball just before the collision is 10m/s downwards. Then the impulse imparted by the platform on the ball is :-



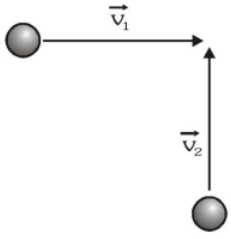
(A) 15 N–s (B) 10 N–s
(C) 20 N–s (D) 30 N–s

Q21. Two identical balls A and B are released from the position shown in Fig. They collide elastically with each other on the horizontal portion. The ratio of heights attained by A and B after collision is(neglect friction).



(A) 1 : 4 (B) 2 : 1
(C) 4 : 13 (D) 2 : 5

Q22. Two balls of masses m_1, m_2 and speeds v_1 and v_2 collide at right angle. The maximum amount of kinetic energy loss due to inelastic collision is ____.



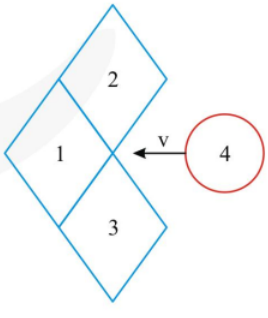
(A) $\frac{1}{2}(m_1v_1^2 + m_2 v_2^2)$

(B) $\frac{1}{2}\frac{m_1m_2}{m_1+m_2}(v_1^2 + v_2^2)$

(C) $\frac{1}{2}\frac{1}{m_1+m_2}(m_1^2v_1^2 + m_2^2v_2^2)$

(D) $\frac{1}{2}\frac{m_1m_2}{m_1+m_2}(v_1+v_2)^2$

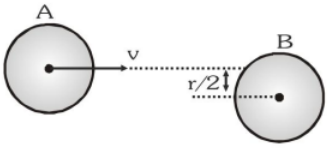
Q23. A smooth washer impinges at a velocity 'v' on a group of three smooth identical blocks resting on a smooth horizontal surface as shown in fig. Mass of each block is equal to mass of the washer. The diameter of the of the washer and its height are equal to edge of the block. The velocity of blocks (2) and (3) after collision is -



(A) v (B) $\frac{v}{\sqrt{2}}$

(C) $\frac{v}{2}$ (D) 2v

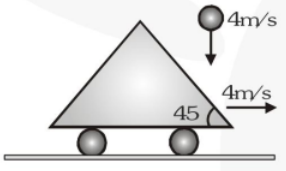
Q24. A disk A of radius r moving on perfectly smooth surface at a speed v undergoes an elastic collision with an identical stationary disk B. Find the velocity of the disk B after collision if the impact parameter is r/2 as shown in the figure :-



(A) $\frac{\sqrt{15}}{4}v$ (B) $\frac{v}{4}$

(C) $\frac{v}{2}$ (D) $\frac{\sqrt{3}v}{2}$

Q25. A small ball falling vertically downward with constant velocity 4m/s strikes elastically a massive inclined cart moving with velocity 4m/s horizontally as shown. The velocity of the rebound of the ball is :-



(A) $4\sqrt{2}$ m/s (B) $4\sqrt{3}$ m/s

(C) 4 m/s (D) $4\sqrt{5}$ m/s

Q26. A spherical ball of mass 1 kg moving with a uniform speed of 1 m/s collides symmetrically with two identical spherical balls of mass 1 kg each at rest touching each other. If the two balls move with 0.5m/s in two directions at the same angle of 60° with the direction of the first ball, the loss of kinetic energy on account of the collision is :-

(A) 0.125 J (B) 0.5J

(C) 1.0 J (D) 0.75J

Q27. A particle moving horizontally collides with a fixed plane inclined at 60° to the horizontal. If it bounces vertically, the coefficient of restitution is :-

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{2}{\sqrt{3}}$

(C) $\frac{1}{3}$ (D) None of these

Q28. A block of mass 2 kg moving with velocity of 20 m/s collide with a block of mass 2 kg attached with one end of a spring. A block of mass 3 kg is attached to the other end as shown. If the collision is elastic then maximum compression in spring will be (k = 1000 N/m)



(A) $20\sqrt{3}$ cm (B) $40\sqrt{3}$ cm

(C) $60\sqrt{3}$ cm (D) None

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A non–zero external force acts on a system of particles. The velocity and acceleration of the centre of mass are found to be v_0 and a_C respectively at any instant t. It is possible that :

(i) $v_0=0$, $a_C=0$

(ii) $v_0 \neq 0$, $a_C=0$

(iii) $v_0= 0$, $a_C\neq0$

(iv) $v_0\neq 0$, $a_C\neq0$

Then -

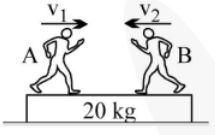
(A) (iii) and (iv) are true.

(B) (i) and (ii) are true.

(C) (i) and (iii) are true.

(D) (ii), (iii) and (iv) are true.

Q2. In the figure shown the system is at rest initially. Two persons 'A' and 'B' of masses 40 kg each move with speeds v_1 and v_2 respectively towards each other on a plank lying on a smooth horizontal surface as shown in figure. Plank travels a distance of 20 m towards right direction in 5 sec. (Here v_1 and v_2 are given with respect to the plank). Then the possible condition(s) can be -



(A) $v_1 = 0$ m/s, $v_2 = 10$ m/s

(B) $v_1 = 5$ m/s, $v_2 = 15$ m/s

(C) $v_1 = 10$ m/s, $v_2 = 20$ m/s

(D) $v_1 = 2$ m/s, $v_2 = 12$ m/s

Q3. Two balls A and B having masses 1 kg and 2 kg, moving with speeds 21 m/s and 4 m/s respectively in opposite direction, collide head on. After collision A moves with a speed of 1 m/s in the same direction, then correct statements is :

(A) The velocity of B after collision is 6 m/s opposite to its direction of motion before collision.

(B) The coefficient of restitution is 0.2.

(C) The loss of kinetic energy due to collision is 200 J.

(D) The impulse of the force between the two balls is 40 N–S.

Q4. A set of n–identical cubical blocks lies at rest parallel to each other along a line on a smooth horizontal surface. The separation between the near surfaces of any two adjacent blocks is L. The block at one end is given a speed v towards the next one at time t = 0. All collisions are completely inelastic, then :-

(A) The last block starts moving at

$$t = n\,(n-1)\frac{L}{2v}$$

(B) The last block starts moving at

$$t = (n-1)\frac{L}{v}$$

(C) The centre of mass of the system will have a final speed $\frac{v}{n}$

(D) The centre of mass of the system will have a final speed v.

Q5. A particle moving with kinetic energy = 3 joule makes an elastic head on collision with a stationary particle which has twice its mass during the impact.

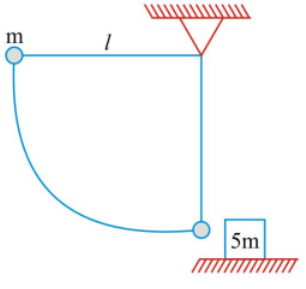
(A) The minimum kinetic energy of the system is 1 joule.

(B) The maximum elastic potential energy of the system is 2 joule.

(C) Momentum and total energy of the system are conserved at every instant.

(D) The ratio of kinetic energy to potential energy of the system first decreases and then increases.

Q6. A pendulum bob of mass m connected to the end of an ideal string of length l is released from rest from horizontal position as shown in Fig. At the lowest point, the bob makes an elastic collision with a stationary block of mass 5m, which is kept on a frictionless surface. Mark out the correct statement(s) for the instant just after the impact.



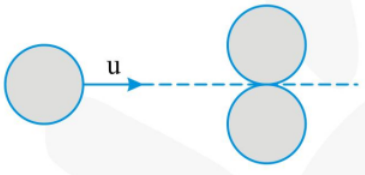
(A) Tension in the string at lowest point just after collision is (17/9) mg.

(B) Tension in the string at lowest point just before collision is 3mg.

(C) The velocity of the block just after collision is $\sqrt{2gl}/3$.

(D) The maximum height attained by the pendulum bob after impact is (measured from the lowest position) $\frac{4l}{9}$.

Q7. Two equal spheres of mass *m* are in contact on a smooth horizontal table. A third identical sphere impinges symmetrically on them and reduces to rest then :



(A) Coefficient of restitution is $e = \frac{2}{3}$

(B) Loss of kinetic energy $\frac{1}{6}mu^2$ where u is velocity before impact

(C) After the collision, velocity of equal mass sphere is $\frac{u}{\sqrt{3}}$

(D) Loss of kinetic energy $\frac{1}{3}mu^2$

Q8. In a one dimensional collision between two identical particles A and B, B is stationary and A has momentum p before impact. During impact, B gives impulse J to A.

(A) The total momentum of the 'A plus B' system is p before and after the impact, and (p-J) during the impact.

(B) During the impact A gives impulse of magnitude J to B.

(C) The coefficient of restitution is $\frac{2J}{p} - 1$.

(D) The coefficient of restitution is $\frac{J}{p} + 1$.

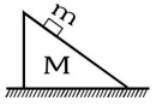
MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q9. In each situation of column–I, a system involving two bodies is given. All strings and pulleys are light and friction is absent everywhere. Initially each body of every system is at rest. Consider the system in all situation of column I from rest till any collision occurs. Then match the statements in column – I with the corresponding results in column–II.

Column I

(A) The block plus wedge system is placed over smooth horizontal surface. After the system is released from rest, the centre of mass of system

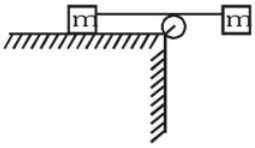


Column II

(P) Shifts towards right

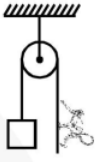
(B) The string connecting both the blocks of mass m is horizontal. Left block is placed over smooth horizontal table as shown. After the two block system is released from rest, the centre of mass of system

(Q) Shifts downwards



(C) The block and monkey have same mass. The monkey starts climbing up the rope. After the monkey starts climbing up, the centre of mass of monkey+block system

(R) Shifts upwards

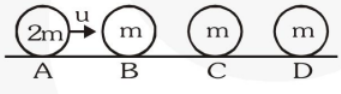


(D) Both block of mass m are initially at rest. The left block is given initial velocity u downwards. Then, the centre of mass of two block system afterwards

(S) Does not shift

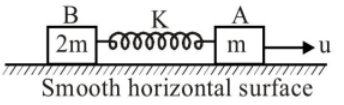


Q10. Four balls A,B,C and D are kept on a smooth horizontal surface as shown in figure. Ball A is given velocity u towards B- (Assume each collision to be elastic)



Column-I	**Column-II**
(A) Total impulse of all collisions on A	(P) $\frac{4mu}{9}$
(B) Total impulse of all collisions on B	(Q) $\frac{4mu}{27}$
(C) Total impulse of all collision on C	(R) $\frac{4mu}{3}$
(D) Total impulse of all collisions on D	(S) $\frac{52}{27}mu$

Q11. Two blocks A and B of mass m and 2m respectively are connected by a massless spring of spring constant K. This system lies over a smooth horizontal surface. At t = 0 the block A has velocity u towards right as shown while the speed of block B is zero, and the length of spring is equal to its natural length at that instant.

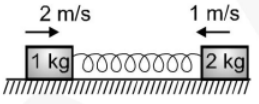


Column-I	Column-II
(A) The velocity of block A	(P) can never be zero
(B) The velocity of block B	(Q) may be zero at certain instants of time
(C) The kinetic energy of system of two block	(R) is minimum at maximum compression of spring
(D) The potential energy of spring	(S) is maximum at maximum extension of spring

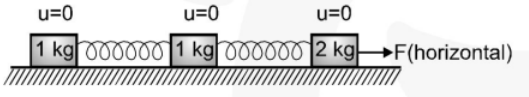
Q12. In all cases in **column–I**, the blocks/plank/trolley are placed on the smooth horizontal surface.

Column–I

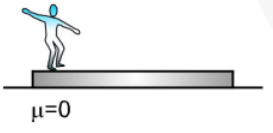
(A) The initial velocities given to the blocks when spring is relaxed are as shown



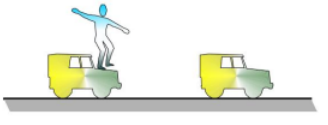
(B) A constant force is applied on 2 kg block. Springs are initially relaxed



(C) Initially system is at rest. Man starts be moving on a large plank with constant velocity.



(D) A man standing on one of the trolleys (initially at rest) jumps to the other with relative velocity of 4 m/s horizontally



Column–II

(P) Centre of mass of the complete system shown will not move horizontally

(Q) Centre of mass of the complete system shown will move horizontally

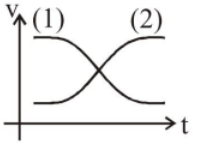
(R) Mechanical energy of the system will be conserved

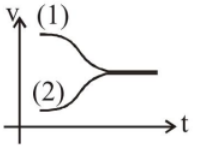
(S) Mechanical energy of the system will increase

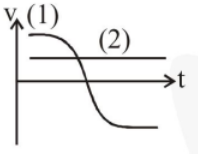
(T) Linear momentum of the complete system will always remain constant

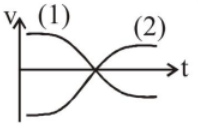
Q13. Assume that 2 bodies collide head on. The graph of their velocities with time are shown in column-I match them with appropriate situation in column-II

Column-I

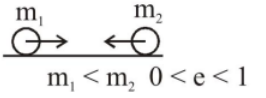
(A) 

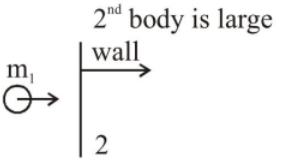
(B) 

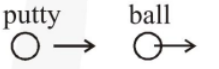
(C) 

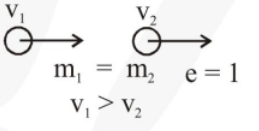
(D) 

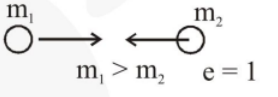
Column-II

(P) 

(Q) 

(R) 

(S) 

(T) 

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 14 to 17)

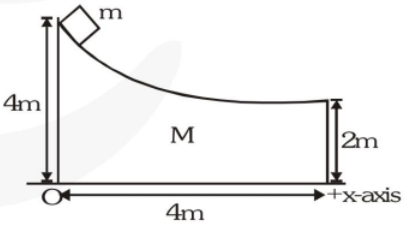
If net force on a system in a particular direction is zero (say in horizontal direction) we can apply:

$$\Sigma m_R x_R = \Sigma m_L x_L,$$

$$\Sigma m_R v_R = \Sigma m_L v_L$$

and $$\Sigma m_R a_R = \Sigma m_L a_L$$

Here R stands for the masses which are moving towards right and L for the masses towards left, x is displacement, v is velocity and a the acceleration (all with respect to ground). A small block of mass m = 1 kg is placed over a wedge of mass M = 4 kg as shown in figure. Mass m is released from rest. All surfaces are smooth. Origin O is as shown.



Q14. Final velocity of the wedge is ..... m/s :-

(A) $\sqrt{3}$ (B) $\sqrt{2}$

(C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{\sqrt{3}}$

Q15. The block will strike the x–axis at x = ...m :-

(A) 4.2 (B) 7.6

(C) 5.6 (D) 6.8

Q16. Normal reaction between the two blocks at an instant when absolute acceleration of m is $5\sqrt{3}\ m/s^2$ at 60° with horizontal is ..... N. Normal reaction at this instant is making 30° with horizontal :-

(A) 6 (B) 10

(C) 4 (D) 5

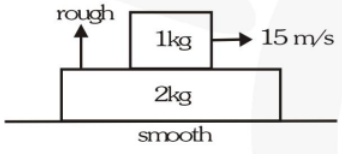
Q17. At the same instant reaction on the wedge from the ground is ..... N.

(A) 42.5 (B) 40

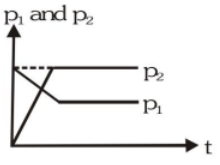
(C) 43.46 (D) None of these

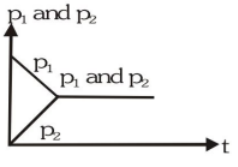
Paragraph # 2 (Ques. 18 to 20)

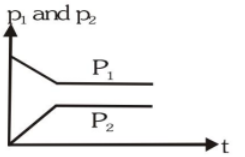
A 1 kg block is given a velocity of 15 m/s towards right over a very long rough plank of mass 2 kg as shown in figure.

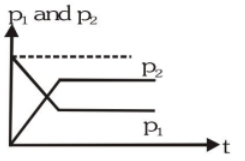


Q18. The correct graph showing linear momentum of 1 kg (i.e. p_1) and of 2kg (i.e. p_2) versus time is :-

(A)


(B)


(C)


(D)


Q19. If coefficient of friction between the two blocks is equal to 0.4, then magnitude of initial slope of p_1 versus t and p_2 versus t (in SI unit) will be :-

(A) 4 and 2 (B) 2 and 4

(C) 4 and 4 (D) 2 and 2

Q20. Momentum of both the blocks are equal at time t = ...seconds :-

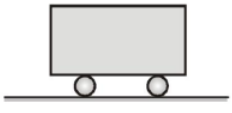
(A) 1.75 (B) 1.875

(C) 2.5 (D) 1.25

Paragraph # 3 (Ques. 21 to 23)

When the mass of a system is variable, a thrust force has to be applied on it in addition to all other forces acting on it. This thrust force is given by :

$\vec{F} = \vec{v}_r\left(\pm\frac{dm}{dt}\right)$. Here $\vec{v}_r$ is the relative velocity with which the mass dm either enters or leaves the system. A car has total mass 50 kg. Gases are ejected from this backwards with relative velocity 20 m/s. The rate of ejection of gas is 2 kg/s.Total mass of gas is 20 kg. Coefficient of friction between the car and road is $\mu = 0.1$.



Q21. Car will start moving after time t = ... second :-

(A) 4 (B) 10

(C) 5 (D) 8

Q22. Maximum speed of car will be v = ... m/s :- (Take $\ell n \frac{4}{3} = 0.28$)

(A) 0.6 (B) 0.8 (C) 1.0 (D) 1.2

Q23. Car will stop after (from starting) t = ... seconds :-

(A) 12.2 (B) 6.4 (C) 10.6 (D) 5.8

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q24. A plank P and block Q are arranged as shown on a smooth table top. They are given velocities 3 m/s and 6 m/s respectively.

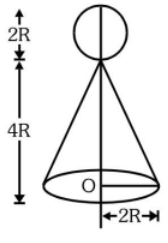
The length of plank is 1m and block is of negligible size. After some time when the block has reached the other end of plank it stops slipping on plank. Find the coefficient of friction between plank P and block Q if mass of plank is double of block.



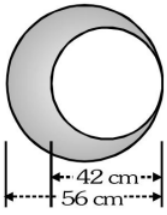
SUBJECTIVE TYPE

Q25. A man has constructed a toy as shown in fig. If density of the material of the sphere is 12 times of the cone compute the position of the centre of mass.

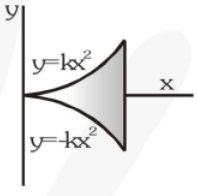
[Centre of mass of a cone of height h is at height of $\frac{h}{4}$ from its base.]



Q26. A circular plate of uniform thickness has a diameter of 56 cm. A circular portion of diameter 42 cm is removed from one edge of the plate as shown in figure. Find the position of the centre of mass of the remaining portion.



Q27. A thin sheet of metal of uniform thickness is cut into the shape bounded by the line x = a and $y = \pm kx^2$, as shown. Find the coefficient of the centre of mass.



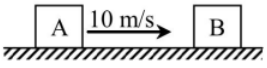
Q28. A spaceship is moving with constant speed v_0 in gravity free space along +Y-axis suddenly shoots out one third of its part with speed $2v_0$ along + X-axis. Find the speed of the remaining part.

Q29. A ball is projected with a velocity u at an angle of elevation α from a point distance d from a smooth vertical wall in a plane perpendicular to it. After rebounding from the wall, it returns to the point of projection.

Prove that $u^2 \sin 2\alpha = gd\left(1+\frac{1}{e}\right)$, where e is coefficient of restitution. Hence find the maximum distance d for which the ball can return to point of projection.

Q30. A sphere of mass m is moving with a velocity $4\hat{i}-\hat{j}$ when it hits a smooth wall and rebounds with velocity $\hat{i}+3\hat{j}$. Find the impulse it receives. Find also the coefficient of restitution between the sphere and the wall.

Q31. The blocks shown in figure have equal masses. The surface of A is smooth but that of B has a friction coefficient of 0.10 with the floor. Block A is moving at a speed of 10m/s towards B which is kept at rest. Find the distance travelled by B if (a) the collision is perfectly elastic and (b) the collision is perfectly inelastic. Take g = 10 m/s^2.



Q32. Two cars initially at rest are free to move in the x direction. Car A has mass 4 kg and car B has mass 2 kg. They are tied together, compressing a spring in between them. When the spring holding them together is burned, car A moves off with a speed of 2 m/s.

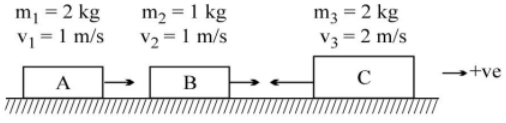
(i) With what speed does car B leave.

(ii) How much energy was stored in the spring before it was burned.

Q33. Three carts move on a frictionless track with inertias and velocities as shown. The carts collide and stick together after successive collisions.

(i) Find loss of mechanical energy when B & C stick together.

(ii) Find magnitude of impulse experienced by A when it sticks to combined mass (B & C).



Q34. A body is thrown vertically upwards from ground with a speed of 10 m/s. If coefficient of restitution of ground, e = 1/2. Find

(i) The total distance travelled by the time it almost stops.

(ii) Time elapsed (after the ball has been thrown) when it is at its subsequent maximum height for the third time.

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
B	C	A	C	B	A	D	B	D	B
11	12	13	14	15	16	17	18	19	20
A	D	C	D	D	C	D	B	A	C
21	22	23	24	25	26	27	28	29	30
C	D	A	A	B	B	B	C	C	C
31	32	33	34	35	36	37			
C	C	A	D	B	D	D			

Ex-1A

1. C,D **2.** A,C **3.** D **4.** A, B,D **5.** A, C

6. B, C

7. (A) → Q; (B) → R; (C) → P

8. (A) → Q, (B) → P,R; (C) → P,R; (D) → R,S

9. (A) → R; (B) → T; (C) → T; (D) → S

10. (A) → Q; (B) → P; (C) → S

11. (A) → Q,T; (B) → R,T; (C) → R,S,T,U **12.** (A)-Q (B)-S (C)-P

13. C **14.** B **15.** B **16.** C **17.** D

18. x = 6m **19.** $\left(\frac{a}{2},\frac{7a}{10}\right)$ **20.** $\sqrt{13}$ m, $\left(\frac{14}{5},\frac{19}{5}\right)$ **21.** $\left(\frac{a}{3},\frac{a}{3}\right)$

22. g/9 downwards **23.** $\frac{7}{18}$

24. **(i)** at O **(ii)** Diagonally from O to 3 **(iii)** Along OY'
(iv) at O **(v)** Diagonally from O to 4 **(vi)** at O

Ex-2

1	2	3	4	5	6	7	8	9	10
D	C	B	A	C	B	C	A	D	C
11	12	13	14	15	16	17	18	19	20
B	A	B	D	C	A	C	B	B	D
21	22	23	24	25	26	27	28		
C	B	B	A	D	A	C	B		

Ex-2A

1. A **2.** A,B,C,D **3.** A,B,C **4.** A,C **5.** A,B,C,D

6. A,B,C,D **7.** A,B,C **8.** B,C

9. (A)-Q; (B)-P, Q; (C)-R; (D) S

10. A-(S), B-(Q), C-(P), D-(R)

11. (A)-Q; (B)-Q; (C)-P,R; (D)-Q,S

12. (A)→(P,R,T); (B)→(Q,S); (C)→(P,S,T); (D)→(P, S, T)

13. (A) S, (B) R, (C) Q, (D) P

14. B **15.** D **16.** D **17.** A **18.** D

19. C **20.** B **21.** C **22.** A **23.** C

24. 0.3 **25.** 4R from O

26. 9 cm from center of bigger circle (leftwards)

27. $x=\frac{3a}{4}$ **28.** $\frac{\sqrt{13}}{2}v_0$ **29.** $d\le\frac{eu^2}{g(1+e)}$ **30.** $m\left(-3\hat{i}+4\hat{j}\right)$, $e=\frac{9}{16}$

31. (a) 50 m (b) 25 m **32.** (i) 4 m/s, (ii) 24 J

33. (i) 3 J, (ii) $\frac{12}{5}$ N-s **34.** (i) $\frac{40}{3}$ m, (ii) t_e = 3.25 s

-: Important Notes & Formulas :-